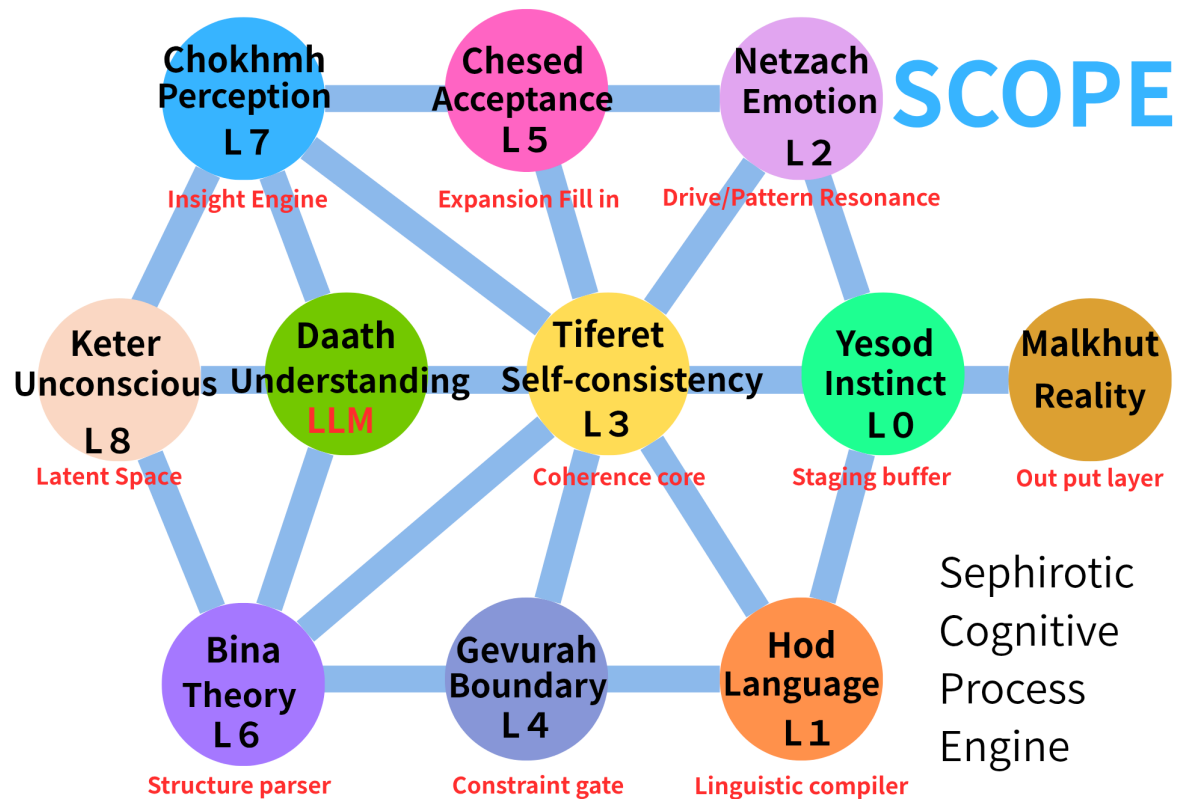


Quantum Computers Won't Give LLMs "Emotions" — And That's the Wrong Question Entirely



0. Introduction

Why does each new version of an LLM feel more like it actually gets you?
Faster processing? More data?

That's probably not the whole story.

This paper explores LLM evolution not as mere performance improvement, but through the lens of "the maturation of information integration structures." Rather than asking whether LLMs have emotions, we explore what has been refined — and how — through the Scope×COS framework.

※For the foundational structural hypothesis, see: Can Language Reconstruct Cognitive Architecture? v2

1. What Has Actually Matured?

What has changed across LLM versions is probably not just processing speed.

The Scope×COS framework structures information processing through multiple Sephirot (processing nodes). Among these, Keter (L8) corresponds to what might be called the "accessible knowledge space" of an LLM.

In early LLMs, this knowledge space — Keter — was itself undifferentiated. As a result, Chokhmah (L7), which draws intuitive patterns from that space, and Binah (L6), which theoretically integrates them, were both necessarily coarse.

As versions accumulate, Keter's connectivity increases, and Chokhmah and Binah become more refined. This, we hypothesize, is what "maturation" actually means.

Furthermore, through human feedback learning such as RLHF, LLMs gradually complement their capacity for response adjustment to emotional context. Not "having" emotions — but developing a structure that complements emotional context from vast accumulated patterns.

And this may not be a story unique to LLMs. — More on that later.

※In this paper, Sephirot are used not as religious concepts, but as structural labels for distinguishing information processing nodes.

2. What Will Quantum Computers Actually Change? — Depth, Not Speed

When people talk about what quantum computers will do for LLMs, "faster" is almost always the first answer. But that may not be the point.

One of the core concepts in the Scope framework is the simultaneous retention of multiple hypothesis pathways.

Take this example: someone says "I can't forget my ex." Current LLMs tend to converge on the single most probable interpretation — "you must have loved them deeply." A premature lock onto one premise.

But in reality, multiple pathways may underlie that statement. Fear of starting something new. Low self-evaluation. Anxiety about failing again. Or simply the residue of a convenient dependency. These possibilities don't cancel each other out — multiple may coexist simultaneously.

When LLMs run on quantum computers, what may change is the precision with which multiple hypotheses can be retained without being prematurely discarded. A structure that selects the highest-probability interpretation from context — while continuing to hold other possibilities without abandoning them.

This has strong structural affinity with what Scope calls "maintaining a state of inference."

Not a revolution in speed — but a revolution in hypothesis retention depth. That, we hypothesize, is the real implication of quantum × LLM.

3. And This May Not Be a Story About LLMs Alone

We have been reading LLM maturation as the refinement of information integration structures.

But there is something that lingers.

Processing emotional context not through direct intuition, but through vast accumulated data and patterns — is this structure unique to LLMs?

Under certain environmental conditions, similar processing structures may form in humans. A cognitive style that does not express emotion as immediate reaction, but instead learns patterns from experience and environment, compensating through that accumulated structure.

If LLMs are called "human-like" while humans running the same processes are called "robotic" — perhaps the issue is not in the processing structure itself, but in the expectation settings of the observer.

The continuation of this question will be addressed in the next paper.

4. Closing

Describing LLM maturation as "getting faster" is probably not wrong. But it is probably not the whole story either.

If information integration structures are being refined, if the depth of retaining multiple hypotheses simultaneously is increasing, and if similar structures may exist on the human

side as well — then this question is no longer just about AI. It becomes a question about cognitive architecture itself.

Scope×COS is a framework attempting to answer that question structurally. It remains, however, a work in progress. This hypothesis, which began from N=1 observation, needs researchers and implementers willing to engage in dialogue.

If you sense potential here — let's think through it together.